



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin

Introduction

CSU44097 Project Methods / CSU44099 Final Year Project

ICS Master's Pathway

Internship (year 4) +
Dissertation (year 5)

ICS Honours Bachelor's Pathway
Joint Honours (bus, ec, geo, ling)
CSLL
Final Year Project

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Research Ethics
Presentation Skills

On successful completion of this module, you will be able to:

Communicate effectively in technical and scientific writing, and present scientific/technical ideas concisely to a technical audience that may not be expert in the specific domain of the project;

Manage workflow and task scheduling within the constraints of the resources available to meet specific design goals and deadlines;

Independently locate and assess potential research sources and evaluate those sources for their relevance to a practical computer science problem, application or topic;

Propose methods of testing and evaluation;

Examine and discuss the impact of specific technology on society, giving consideration to ethical norms and standards.

general computer science audience

your time, your supervisor's time, people you need to collaborate with, resources you need access to, experiment participants ...

background, related work
relationship to your project
your literature review

to answer your research question(s)
might require ethics approval ...

... following an approval process that requires you to think carefully about the design and conduct of your study

OR

Week 1

Introduction for students completing a Final Year Project (exiting after Year 4)

Week 2

Introduction for students completing an Internship followed by a Dissertation next year (exiting after Year 5)

Weeks 3 & 4

(**Lucy Hederman, Jonathan Dukes, Aoife O'Connor**)

Planning your report writing, conducting your literature review, using library databases and referencing tools

Week 5 (Jonathan Dukes)

Planning your project, managing your time, identifying, quantifying and mitigating risks

Week 6 (TBC)

Research ethics and the ethics approval process (REAMS)

Week 7

STUDY WEEK

Week 8 (Tim Savage)

Gender and inclusivity in research

Weeks 9 & 10 (Emma Louise Howard)

Evaluating and presenting your results

Week 11 (TBC)

Presenting your work

Week 12

Reserved for assessment

CSU44097 – Project Methods (5 credits)

Integrated Computer Science (exiting after Year 5)

Project Methods is a standalone module, separate from your internship and dissertation

Project Methods is assessed separately from your internship and dissertation

You pass/fail **Project Methods** separately

CSU44099 – Final Year Project (20 credits)

Integrated Computer Science (exiting after Year 4)

Joint Honours Computer Science with Business, Economics, Linguistics or Geography

Computer Science, Linguistics and a Language

Project Methods is a component of your Final Year Project module – not a separate module



IMPORTANT NOTE: From 2025/2026 onwards, after Week 2 of Semester 1, you cannot switch from the 5-year MCS pathway to the 4-year BA (Mod) pathway and exit in the same year with an honours BA (Mod) degree (See Handbook section 3.14)

To switch from Master in Computer Science (MCS, 5-year) to BA Mod (4-year pathway) you must do one of the following:

Request the change **before the end of Week 2 of Semester 1** (teaching-unit@rt.scss.tcd.ie)
A Final Year Project supervisor will be assigned to you if you do not already have one

Only way to graduate in
with an honours BA (Mod)
degree this year

OR

Complete Semester 1 of Year 4 plus 10 credits in Semester 2 (options *) **and defer your Final Year Project to the next academic year** (which would have been Year 5)

or exit immediately with an Ordinary BA degree **reflecting the first three years of study only (not classified First, Two-One, Two-Two, etc.)**

You cannot exit with an honours BA Mod degree (or a Master's in Computer Science degree) without completing a capstone project. The internship is not considered a replacement for a capstone project.

See Section 3.14 of your ICS course handbook for more details

* There are two options available to accumulate 10 credits in Semester 2 – see course handbook

CSU44097 – Project Methods (5 credits)

Integrated Computer Science (exiting after Year 5)

Assessment

20%

MCQ Test

40%

Research Ethics
Assignment

40%

Project Presentation
Video

CSU44099 – Final Year Project (20 credits)

Integrated Computer Science (exiting after Year 4)

Joint Honours Computer Science with Business,
Economics, Linguistics or Geography

Computer Science, Linguistics and a Language

Assessment

5%

MCQ Test

10%

Interim Final Year
Project Report

0%

Final Year Project
Presentation

85%

Final Year Project
Report

Looking ahead to the Computer Science Dissertation

What is the MCS Dissertation?

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Submitted in Year 5 of the Integrated CS Master's pathway

A substantial piece of individual **academic writing**

Typically 15,000–20,000 words or 50–70 pages

Describing an individual research project

30 credits, approx 600-750 hours (approx 20–25 hours per week) over 30 weeks

No taught component (lectures) specific to your topic

some supporting skills and guidance

provided this year here in Project Methods

Your supervisor (next year) will provide guidance and feedback

but you will work with much more independence than you have before
(and more than you will have during your internship)

Only the dissertation itself counts towards your mark for 30 credits

If it is not in your dissertation, then it never happened

describing your project clearly and concisely through your dissertation is essential

your supervisor's impression of the conduct of the project is not directly considered

your code (if you wrote any) is not usually considered

Double Marking

Independently marked by (1) your supervisor and (2) a “Second Reader”



Mark independently without consultation and using the same scheme (“rubric”)

Marks agree within 15%: supervisor and second reader discuss and agree final mark

Marks differ by > 15%: “Third Reader” is assigned, will independently assess the dissertation and discuss a final mark with the supervisor and second reader



Second Reader

Assume the second reader is not an expert on the topic of your project

e.g. might be expert in graphics but your project is on networks

Second Reader’s first sight of your project will be during your project presentation

During the presentation, they might (or might not) give you some suggestions for the dissertation

Finding a Dissertation Supervisor

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August/
September
2027

<https://projects.scss.tcd.ie/projects/>

Your first and most important task in August/September 2027!!

Projects will be proposed on <https://projects.scss.tcd.ie>

narrow your search by **project area** (keywords), **discipline**, or **supervisor**

proposals can often be adapted, sometimes into more than one project

don't assume the project is "set in stone"

can propose your own project to a supervisor working in a related area

You will contact the supervisor(s) of your preferred project(s) by email (hint: consider contacting 2 or 3 potential supervisors)

supervisors may wish to discuss any proposed project with you

and you might have questions for them

Recommendation: Contact preferred supervisor(s) 1–2 weeks before teaching starts – timing this right is important!!

Can I frame two or three research questions related to the topic?

Can I perform the work (e.g. experiments) to answer those questions in the time available?

What might a schedule for the project look like?

Does the topic lend itself to writing a literature review that explores related published research?

Would I need ethics approval to conduct my experiments? Am I likely to obtain approval?

I would put this in my diary **now!!**

Each supervisor will have their own approach to managing projects

Time with your supervisor will be limited and valuable

as a guide, expect to see them for 30 minutes every 1–2 weeks
(if you miss a meeting, it might not be possible to reschedule)

set aside time in your weekly schedule to prepare for your supervisor meetings
(find out whether they expect to receive anything in advance of meetings)

discuss whether (and when) they can provide you with feedback on your deliverables
(draft interim report, draft chapters for your final report)

they normally won't be able to assist you directly with technical issues
(you might be directed to other supports or resources)

Semester 1

refine the scope and aims of the project

review relevant literature and documents
(published journal articles, conference papers,
technical documents)

draft research questions

develop a project work plan

consider whether evaluation methods will require
research ethics approval (and apply for approval)

perform the technical ground work
(identify and fill gaps in your knowledge)

identify long lead-in times

begin the core work of the project
(development, experimentation, ...)

begin drafting interim report
(including literature review extract)

Semester 2

complete and submit interim report

continue with the core work of the project
(development, experimentation, ...)

draft and deliver presentation
(to supervisor and *second reader*)

write and submit final report
(consider viewing this as a full-semester or
even full-year activity)

Your supervisor will be your main point of contact in relation to the content and conduct of **your project**

might not know when your deliverables are due,
or how to submit them,
or who to submit them to,
or how they are weighted and marked, ...

Your course director will be your main point of contact for details that apply to **all projects**

will not know about the specifics of your project

Contact your Course Director with concerns about project supervision

we can deal with general questions here

Your Internship

(and how it is supported by “Project Methods”)

In-Company Presentation [10%]

Technology Management Report [30%]

Consider asking your supervisor for feedback on a draft

Description of the tools and processes

Critical analysis of the strengths and weaknesses of the approaches

Design Report [40%]

Consider asking your supervisor for feedback on a draft

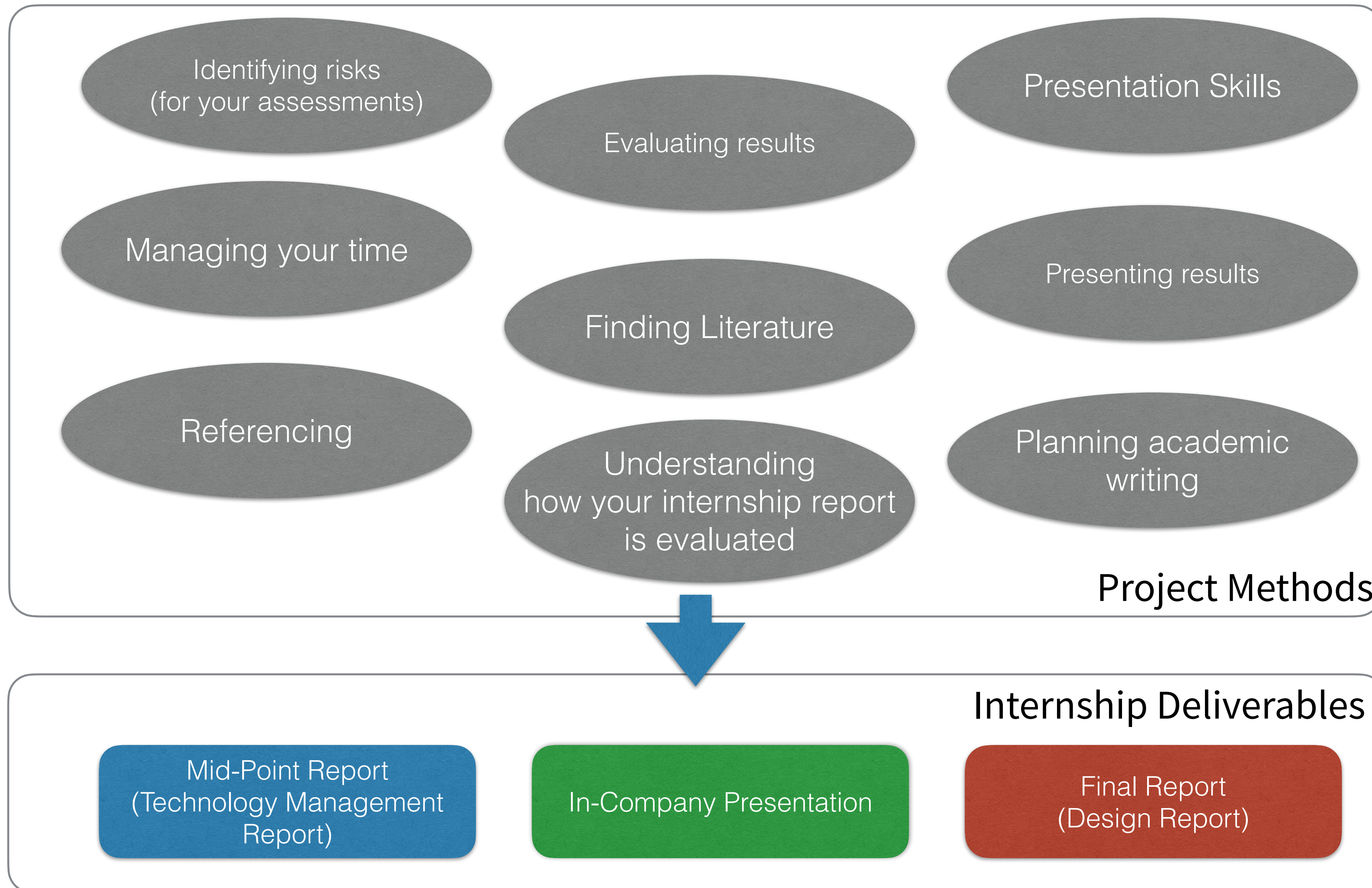
Executive summary

Detailed design report(s) including your independent contribution

Analysis and Conclusions

“... the technical report should **demonstrate a level of individual scholarship** beyond the level of simply fulfilling your role in the organisation. To take a simple example, if your role involves using a particular technology, you could investigate and describe—from an academic perspective—alternative technologies to those used.”

“The report should provide a detailed description of the work you undertook during your internship, both in terms of any technical artefacts you produced, and the organisation in which you participated. With this in mind, **your report should introduce technical concepts using background reference material - cite technical reports, papers, books or documents which might help illustrate or understand what specifically you were doing.**”



Watch the Video on the “Nature of Research and Inquiry” (Tim Savage)

Finally, a few suggestions based on my personal preferences ...

Explore Overleaf – an online LaTeX editor (<https://www.overleaf.com>)

hint: link it with your TCD email address to gain access to premium features

hint: look for a TCD dissertation template

Explore Zotero – an online or app-based reference manager (<https://www.zotero.org/>)

hint: check out how to link Zotero with Overleaf

or: check out the new Library feature in Overleaf

Questions?
Concerns?

See you **next Week (and every week afterwards)**